

Marine Temporal-Evidence Integration Preview

A. Temporal evidence in simulated surface navigation

We apply the same evidence-contract principle to simulated surface navigation, where navigation-software success and interval-qualified physical task completion are distinct propositions. Removal-only evidence conditions distinguish reported status, instantaneous measurements, and sampled post-result behavior, preserving unknown conditions rather than upgrading them into successful docking. Marine-specific adapters and computations implement the paper’s truth-versus-support distinction; this is not zero-shot transfer of an unchanged land implementation.

The application contract fixes a five-second dwell beginning at the first delivered post-result odometry observation. Requirements are position error at most 0.40 m, wrapped heading error at most 0.35 rad, measured translational speed at most 0.05 m/s, absolute yaw rate at most 0.05 rad/s, an oriented hull within a declared berth, and absence of prohibited contact. Nav2’s stopped goal checker evaluates pose and instantaneous velocity conditions; action success does not certify this additional dwell or contact requirement [1]. A mismatch with the stricter application contract is not evidence of a navigation-stack bug.

We fixed the entire closed development cohort before independently recomputing descriptive findings from raw delivered odometry and declared contracts (Table I). Its 48 attempts comprise 24 geometry draws with two tolerance variants each; 22 pairs are fully admitted. All 46 admitted recordings contain an identified action-success receipt. Forty-five cover the sampled dwell under the declared 0.06 s maximum-gap rule. Twenty-three contain a required-condition violation inside the dwell; 22 satisfy observed kinematic/hull checks over the sampled interval but lack contact coverage; one has incomplete coverage and no observed violation. Contact is unknown throughout. These are retained-development counts, not population prevalence, a navigation-error rate, or explanation accuracy.

Figure 1 links one timeline to its evidence ladder. The result-adjacent position error is 0.3982 m and measured speed is 0.0438 m/s. A within-dwell sample at simulator time 284.7148 s has error 0.400264 m, exceeding the bound. L0 supports reported success; L1 adds bounded adjacent measurements; L2 supplies an interval-violation witness. The plot anchors at the first post-result observation, not an invented simulator timestamp for the client receipt. Sampled satisfaction cannot prove continuous

TABLE I

Closed marine development cohort. The final three rows partition admitted recordings; preceding rows are flow counts or marginals.

Flow or evidence category	Records
Scheduled / admitted / technical failures	48 / 46 / 2
Identified software-success receipt	46
Complete sampled dwell coverage	45
Within-dwell violation	23
Sampled compliance; contact unknown	22
Incomplete dwell; no observed violation	1

compliance without intersample bounds. Motion does not uniquely identify waves, current, wind, or actuator defects [2].

Separate inspected B2/B4 comparisons tied useful coverage: both communicated all 24 common units and both answerable sampled-compliance intervals. Strict 5/6 versus 6/6 contact-replay scores hinge on a roughly 0.000145 m temporal-reference discrepancy and do not establish material superiority. Recent deterministic-output checks establish numerical/contract conformance, not independently judged language correctness. Existing docking-policy explanations use surrogate models [3]; our case concerns support for execution completion. It establishes a reproducible application of the evidence-calibration principle and descriptive simulator behavior, while comparative effectiveness, continuous-time proof, and hardware transfer remain unestablished.

References

- [1] Open Navigation LLC, “Nav2 jazzy controller server source, commit f4108e5,” 2026. [Online]. Available: https://github.com/ros-navigation/navigation2/blob/f4108e5b1c2bce804a1aa0c7be6673a8eb4a1501/nav2_controller/src/controller_server.cpp
- [2] T. I. Fossen, Handbook of Marine Craft Hydrodynamics and Motion Control, 2nd ed. Wiley, 2021. [Online]. Available: <https://www.fossen.biz/html/marineCraftModel.html>
- [3] V. B. Gjørsum, I. Strömke, O. A. Alsos, and A. M. Lekkas, “Explaining a deep reinforcement learning docking agent using linear model trees with user adapted visualization,” Journal of Marine Science and Engineering, vol. 9, no. 11, p. 1178, 2021. [Online]. Available: <https://arxiv.org/abs/2203.00368>

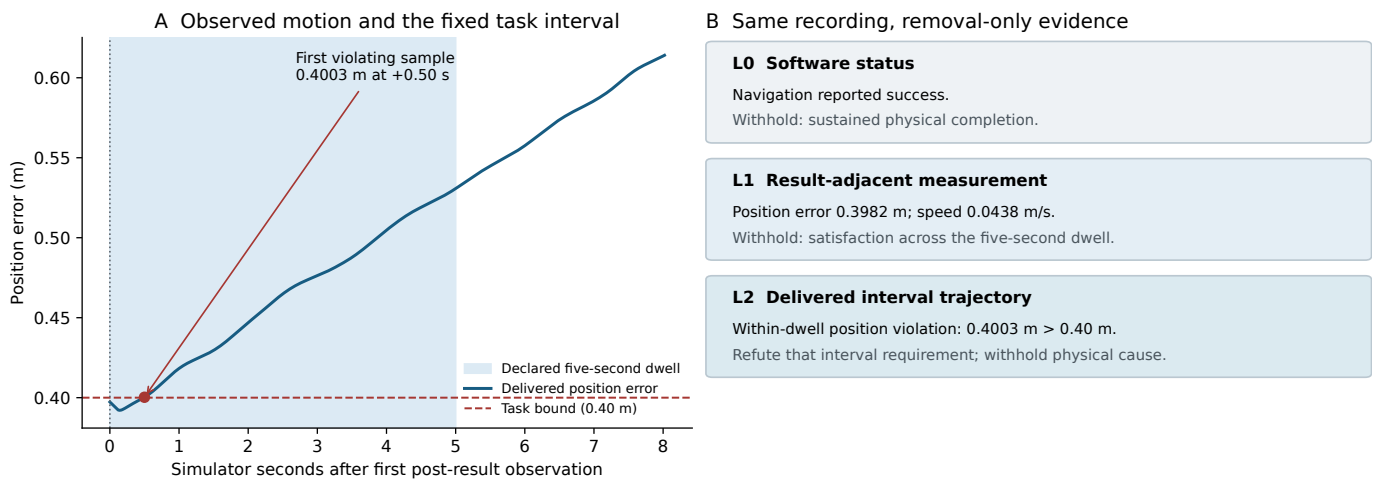


Fig. 1. One retained simulator recording under removal-only evidence. Shading marks the fixed dwell; the point is the first position violation. Time zero is the first delivered post-result observation, not an aligned client-receipt timestamp. Lines connect observed samples for display, without proving intersample compliance. The illustrative record is the first eligible scheduled position-violation case with complete sampled coverage; no representative-frequency or method-score claim follows.